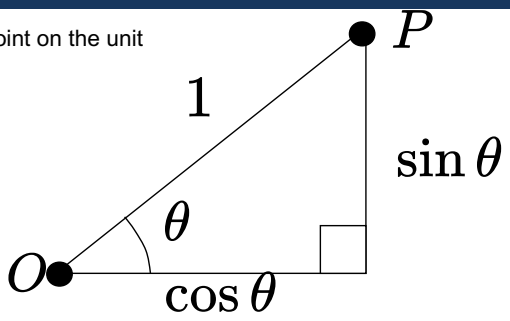
## 1.42 Trigonometry: Trigonometric Identities in Degrees

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### Trigonometric Identities

12

Returning to our point on the unit circle…




1

2 Pythagoras gives you...

$\sin^2\theta$ is a shorthand for $(\sin\theta)^2$. It does NOT mean the sin is being squared – this does not make sense as sin is a function, and not a quantity that we can square!

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### Application of identities #1: Proofs

13

Prove that $1 - \tan\theta\sin\theta\cos\theta \equiv \cos^2\theta$

**Tip #1**: Turn any tan's into sin's and cos's.

$$\tan x = \frac{\sin(x)}{\cos(x)}$$
$$\sin^2 x + \cos^2 x = 1$$

Recall that $\equiv$ means 'equivalent to', and just means the LHS is **always** equal to the RHS for all values of $\theta$.
From Chapter 7 ('Proofs') we saw that usually the best method is to manipulate one side (e.g. LHS) until we get to the other (RHS).

## 1.42 Trigonometry: Trigonometric Identities in Degrees

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### More Examples

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Prove that $\tan\theta + \frac{1}{\tan\theta} \equiv \frac{1}{\sin\theta\cos\theta}$

Simplify $5 - 5\sin^2\theta$

**Tip #2**: In any addition/subtraction involving at least one fraction (with trig functions), always combine algebraically into one.

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### Test Your Understanding

15

Prove that $\frac{\tan x \cos x}{\sqrt{1-\cos^2 x}} \equiv 1$

Prove that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} \equiv 1 - \tan^2\theta$

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Prove that $\tan^2\theta \equiv \frac{1}{\cos^2\theta} - 1$

## 1.42 Trigonometry: Trigonometric Identities in Degrees

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### Exercise 10C – Y1 Page 211-212

16

**10** In each of the following, eliminate $\theta$ to give an equation relating $x$ and $y$:

**a** $x = \sin\theta, y = \cos\theta$ **b** $x = \sin\theta, y = 2\cos\theta$

**c** $x = \sin\theta, y = \cos^2\theta$ **d** $x = \sin\theta, y = \tan\theta$

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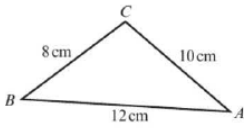
### Exercise 10C – Y1 Page 211-212

17

**11** The diagram shows the triangle $ABC$ with $AB = 12\,\text{cm}$, $BC = 8\,\text{cm}$ and $AC = 10\,\text{cm}$.

**a** Show that $\cos B = \frac{9}{16}$. \hfill (3 ma

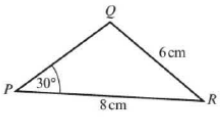
**b** Hence find the exact value of $\sin B$.




**12** The diagram shows triangle $PQR$ with $PR = 8\,\text{cm}$, $QR = 6\,\text{cm}$ and angle $QPR = 30°$.

**a** Show that $\sin Q = \frac{2}{3}$.

**b** Given that $Q$ is obtuse, find the exact value of $\cos Q$.



## 1.42 Trigonometry: Trigonometric Identities in Degrees

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### Quadratics in sin/cos/tan

29

We saw that an equation can be 'quadratic in' something, e.g. $x - 2\sqrt{x} + 1 = 0$ is 'quadratic in $\sqrt{x}$', meaning that $\sqrt{x}$ could be replaced with another variable, say $y$, to produce a quadratic equation $y^2 - 2y + 1 = 0$.

Solve $5\sin^2 x + 3\sin x - 2 = 0$ in the interval $0 \leq x \leq 360^\circ$.

Method 1: Use a substitution.

Method 2: Factorise without substitution.

This is the same, but we 'imagine' $\sin x$ as a single variable and hence factorise immediately.

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### More Examples

30

Solve $\tan^2\theta = 4$ in the interval $0 \leq x \leq 360^\circ$.

Missing the negative case would result in the loss of multiple marks. Beware!

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ in the interval $-180^\circ \leq x \leq 180^\circ$.

**Tip**: We have an identity involving $sin^2$ and $cos^2$, so it makes sense to change the squared one that would match all the others.

## 1.42 Trigonometry: Trigonometric Identities in Degrees

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### Test Your Understanding

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**(a)** Show that the equation $5\sin x = 1 + 2\cos^2 x$ can be written in the form $2\sin^2 x + 5\sin x - 3 = 0$. (2)
**(b)** Solve, for $0 \leq x < 360^\circ$, $2\sin^2 x + 5\sin x - 3 = 0$. (4)

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### Exercise 10F – Y1 P221-222

32

**5** Find all the solutions, in the interval $0 \leq x \leq 360^\circ$, to the equation $8\sin^2 x + 6\cos x - 9 = 0$ giving each solution to one decimal place. (6 marks)

**6** Find, for $0 \leq x \leq 360^\circ$, all the solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ giving each solution to one decimal place.

## 1.42 Trigonometry: Trigonometric Identities in Degrees

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### Exercise 10F – Y1 P221-222

33

**7** Show that the equation $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

**8 a** Show that the equation $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

**b** Hence show that the equation $\cos^2 x = 2 - \sin x$ has no solutions. (3 marks)

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### Exercise 10F – Y1 P221-222

34

**9**\quad $\tan^2 x - 2\tan x - 4 = 0$.
**a** Show that $\tan x = p \pm \sqrt{q}$ where $p$ and $q$ are numbers to be found.
**b** Hence solve the equation $\tan^2 x - 2\tan x - 4 = 0$ in the interval $0 \leq x \leq 540^\circ$.